Ordinary Differential Equations

A Reference List of Types and Techniques

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How to use this document

The central skill in solving ODEs is *recognition*: before reaching for a method, identify the structural type of the equation. Each section below names a type, states the recognition cue, and gives worked examples. The sections are ordered so that each technique builds naturally on the previous ones.

The document is divided into two parts. Part I covers closed-form methods: you recognise a pattern and obtain an elementary function as the answer. Part II covers series methods, which you reach when Part I offers no foothold.

Part I

Closed-Form Methods

1 Separable equations

Recognition. The equation can be written as

$$f(y)\,dy = g(x)\,dx,$$

i.e. the right-hand side factors into a function of x alone times a function of y alone.

Method. Separate and integrate both sides independently.

Examples.

$$\frac{dy}{dx} = xy^2, \qquad \frac{dy}{dx} = \frac{1+y^2}{1+x^2}, \qquad \frac{dy}{dx} = y\cos x.$$

2 First-order linear equations (integrating factor)

Recognition. The equation is linear in y and y':

$$y' + P(x)\,y = Q(x).$$

Method. Multiply through by the integrating factor $\mu(x) = e^{\int P(x)\,dx}$. The left side becomes $(\mu y)'$; integrate.

Examples.

$$y' + 2y = e^{-x}, \qquad y' + \frac{y}{x} = x^2, \qquad y' - y\tan x = \sin x.$$

3 Exact equations

Recognition. Written as $M(x,y)\,dx + N(x,y)\,dy = 0$, and

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

Method. Find F satisfying $F_x = M$ and $F_y = N$. The solution is $F(x,y) = C$.

Examples.

$$(2xy+3)\,dx + (x^2-1)\,dy = 0, \qquad e^y\,dx + (xe^y + 2y)\,dy = 0.$$

4 Non-exact equations made exact (integrating factor for forms)

Recognition. The equation $M\,dx + N\,dy = 0$ is not exact, but

$$\frac{M_y - N_x}{N} \text{ depends only on } x \quad \left(\text{or } \frac{N_x - M_y}{M} \text{ on } y\right).$$

Method. Compute μ from the ratio above and multiply through to make the equation exact.

Example.

$$(2y^2 + 3x)\,dx + 2xy\,dy = 0.$$

5 Homogeneous equations (substitution $v = y/x$)

Recognition. Both sides are homogeneous of the same degree, or equivalently $dy/dx = F(y/x)$ for some function F.

Method. Substitute $y = vx$, so $y' = v + xv'$. The equation becomes separable in v and x.

Examples.

$$\frac{dy}{dx} = \frac{x+y}{x}, \qquad (x^2+y^2)\,dx - 2xy\,dy = 0.$$

6 Bernoulli equations

Recognition. The equation has the form

$$y' + P(x)\,y = Q(x)\,y^n, \quad n \neq 0, 1.$$

It is nonlinear because of the y^n term on the right.

Method. Substitute $v = y^{1-n}$ to obtain a linear equation in v.

Examples.

$$y' + \frac{y}{x} = xy^2, \qquad y' + y = y^3.$$

7 Substitution $v = ax + by + c$

Recognition. The right-hand side depends only on the linear combination $ax + by + c$:

$$\frac{dy}{dx} = f(ax + by + c).$$

Method. Set $v = ax + by + c$. Then $v' = a + bf(v)$, which is separable.

Examples.

$$\frac{dy}{dx} = (x + y + 1)^2, \qquad \frac{dy}{dx} = \cos(x + y).$$

8 Second-order linear, constant coefficients, homogeneous

Recognition. The equation is

$$ay'' + by' + cy = 0, \quad a, b, c \in \mathbb{R},\ a \neq 0.$$

Method. Solve the characteristic equation $ar^2 + br + c = 0$ and apply the appropriate case.

Case 1: distinct real roots $r_1 \neq r_2$

$$y'' - 5y' + 6y = 0 \quad (r = 2, 3) \ \Rightarrow\ y = C_1 e^{2x} + C_2 e^{3x}.$$

Case 2: repeated root $r_1 = r_2 = r$

$$y'' - 4y' + 4y = 0 \quad (r = 2) \ \Rightarrow\ y = (C_1 + C_2 x)e^{2x}.$$

Case 3: complex conjugate roots $r = \alpha \pm \beta i$

$$y'' + 2y' + 5y = 0 \quad (r = -1 \pm 2i) \ \Rightarrow\ y = e^{-x}(C_1 \cos 2x + C_2 \sin 2x).$$

9 Method of undetermined coefficients

Recognition. The equation is $ay'' + by' + cy = g(x)$ where g is a polynomial, exponential, sine/cosine, or a product of these.

Method. Guess a particular solution y_p of the same form as g. If any term of the trial y_p solves the homogeneous equation, multiply the trial form by x (or x^2, etc.) to avoid resonance.

Examples.

$$y'' - 3y' + 2y = e^{3x}, \qquad y'' + y = \sin 2x, \qquad y'' - y = x^2.$$

Resonance example: $y'' + y = \cos x$ — the trial form $A \cos x$ solves the homogeneous equation, so use $y_p = x(A \cos x + B \sin x)$ instead.

10 Variation of parameters

Recognition. The equation is $y'' + P(x)y' + Q(x)y = g(x)$, and undetermined coefficients is not applicable because g is not of the required form (e.g. $g = \sec x$, $\tan x$, $\ln x$).

Method. Let y_1, y_2 be a fundamental pair of solutions to the homogeneous equation with Wronskian W. Then

$$y_p = -y_1 \int \frac{y_2\, g}{W}\, dx + y_2 \int \frac{y_1\, g}{W}\, dx.$$

Examples.

$$y'' + y = \sec x, \qquad y'' - 2y' + y = \frac{e^x}{x}.$$

11 Cauchy–Euler (equidimensional) equations

Recognition. The coefficient of $y^{(k)}$ is a constant multiple of x^k:

$$x^2y'' + \alpha xy' + \beta y = g(x).$$

Method. Try $y = x^r$. This yields a quadratic indicial equation in r, with the same three-case structure as the constant-coefficient characteristic equation. Alternatively, substitute $x = e^t$ to convert to a constant-coefficient equation in t.

Examples.

$$x^2y'' + 2xy' - 6y = 0, \qquad x^2y'' - 3xy' + 4y = x^2.$$

12 Reduction of order

Recognition. A second-order linear ODE for which one solution y_1 is already known.

Method. Set $y = v(x)\, y_1(x)$. Substituting reduces the order by one (the v term drops out), yielding a first-order linear equation in v'.

Example.

$$x^2y'' - xy' + y = 0, \quad y_1 = x.$$

13 Missing dependent variable ($y'' = f(x, y')$)

Recognition. The equation does not contain y explicitly.

Method. Set $p = y'$. Then $y'' = p'$, reducing to a first-order equation in $p(x)$.

Example.

$$y'' = x + y'.$$

14 Missing independent variable ($y'' = f(y, y')$)

Recognition. The equation does not contain x explicitly.

Method. Set $p = y'$ and write

$$y'' = \frac{dp}{dx} = \frac{dp}{dy}\frac{dy}{dx} = p\,\frac{dp}{dy}.$$

This yields a first-order equation in $p(y)$.

Examples.

$$y\, y'' + (y')^2 = 0, \qquad y'' + (y')^2 = 0.$$

Part II

Series Methods

These methods apply when Part I offers no foothold: variable coefficients that are not equidimensional, no known closed-form solution, no obvious second solution for reduction of order.

Structural parallel. The indicial equation in the Frobenius method plays the same role as the characteristic equation in constant-coefficient ODEs. Its two roots r_1, r_2 determine solution form, and the three cases below mirror exactly the three root cases of Section 8.

Constant-coefficient char. eq.	**Frobenius indicial eq.**
Distinct real roots	$r_1 - r_2 \notin \mathbb{Z}$: two clean Frobenius series
Repeated root	$r_1 = r_2$: one series, second solution has $\ln x$
Complex conjugate roots	$r_1 - r_2 \in \mathbb{Z}_{>0}$: second solution *may* have $\ln x$

The repeated-root logarithm arises in Frobenius for the same reason a factor of x appears in the constant-coefficient case: a dimension of solution space must be manufactured by differentiating with respect to the parameter r.

15 Power series at an ordinary point

Recognition. Write the equation as $y'' + p(x)y' + q(x)y = 0$. If p and q are analytic at x_0 (e.g. polynomial coefficients with leading coefficient nonzero at x_0), then x_0 is an *ordinary point.* Two linearly independent power series solutions converge in a neighbourhood.

Method. Substitute $y = \sum_{n=0}^{\infty} a_n(x - x_0)^n$, collect powers of $(x - x_0)$, and derive a recurrence relation for a_n.

Warm-ups (recovery of known functions)

These confirm the bookkeeping before meeting genuinely new functions.

$$y' = y \;\Rightarrow\; y = \sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x.$$

$$y'' + y = 0 \;\Rightarrow\; \text{two series recovering } \cos x \text{ and } \sin x.$$

Airy equation

$$y'' - xy = 0.$$

The simplest equation whose solutions are genuinely non-elementary. The recurrence links every third coefficient; the two independent solutions are the Airy functions $\mathrm{Ai}(x)$ and $\mathrm{Bi}(x)$. Work this one by hand at least once.

Hermite equation

$$y'' - 2xy' + 2\lambda y = 0.$$

When $\lambda = n$ is a non-negative integer, one of the two series terminates, producing the Hermite polynomial $H_n(x)$. The “termination at integer parameter” pattern recurs throughout classical ODEs.

Legendre equation

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0.$$

Solved around the ordinary point $x_0 = 0$. One series terminates when n is a non-negative integer, giving the Legendre polynomials $P_n(x)$. Note that $x = \pm 1$ are singular points, making them natural candidates for Part B below.

Chebyshev equation

$$(1-x^2)y'' - xy' + \lambda^2 y = 0.$$

Same singular-point structure as Legendre. Polynomial solutions arise when $\lambda = n$.

16 Frobenius method at a regular singular point

Recognition. The point x_0 is singular (the leading coefficient vanishes there), but

$$(x-x_0)\,p(x) \quad \text{and} \quad (x-x_0)^2\,q(x)$$

are both analytic at x_0. Such a point is called a *regular* singular point.

Method. Substitute $y = (x-x_0)^r \sum_{n=0}^{\infty} a_n (x-x_0)^n$ with $a_0 \neq 0$. The lowest power of $(x-x_0)$ yields the **indicial equation**

$$r(r-1) + p_0\,r + q_0 = 0,$$

where $p_0 = \lim_{x\to x_0}(x-x_0)p(x)$ and $q_0 = \lim_{x\to x_0}(x-x_0)^2 q(x)$. Denote the roots $r_1 \geq r_2$.

Case 1: $r_1 - r_2 \notin \mathbb{Z}$

Both roots give independent Frobenius series with no logarithms.

Example.

$$2x\,y'' + y' + y = 0.$$

Indicial equation: $r(2r-1) = 0$, roots $r_1 = \frac{1}{2}$, $r_2 = 0$. Difference $\frac{1}{2}$, not an integer. Two clean series follow.

Case 2: $r_1 = r_2$ (repeated root)

One Frobenius series comes from r_1. The second solution necessarily involves a logarithm:

$$y_2 = y_1 \ln x + x^{r_1} \sum_{n=1}^{\infty} b_n x^n.$$

Example: Bessel equation of order zero.

$$x^2 y'' + xy' + x^2 y = 0.$$

Indicial equation: $r^2 = 0$, so $r_1 = r_2 = 0$. The first solution is $J_0(x)$. The second solution $Y_0(x)$ contains $J_0(x)\ln x$ plus a Frobenius-type series.

Case 3: $r_1 - r_2 \in \mathbb{Z}_{>0}$ (integer-differing roots)

The larger root r_1 always produces a Frobenius series. The smaller root r_2 *may or may not* work: if the recurrence breaks down, the second solution requires a logarithm; if it does not break down, two clean series exist.

Example A (no logarithm):

$$xy'' + 2y' + xy = 0,$$

indicial roots $r = 0, -1$ (differ by 1). The smaller root happens to work; two series are obtained without a logarithm.

Example B (logarithm required): Bessel equation of order one.

$$x^2y'' + xy' + (x^2 - 1)y = 0,$$

indicial roots $r = \pm 1$ (differ by 2). The smaller root fails in the recurrence; the second solution $Y_1(x)$ requires a logarithmic term.

Comparing Examples A and B side-by-side is the best way to internalise why Case 3 is the subtle one.

17 Suggested practice order

Part I (closed-form methods)

Separable $\to$ linear (integrating factor) $\to$ exact $\to$ homogeneous substitution $\to$ Bernoulli $\to$ second-order constant coefficient (all three root cases) $\to$ undetermined coefficients (include a resonance example) $\to$ Cauchy–Euler $\to$ variation of parameters $\to$ reduction of order $\to$ missing-variable substitutions.

Part II (series methods)

Sanity-check series (e^x, then $\sin x$ and $\cos x$) $\to$ Airy equation (first genuinely new function) $\to$ Frobenius Case 1 ($2xy'' + y' + y = 0$) $\to$ Frobenius Case 2 (Bessel order 0) $\to$ Frobenius Case 3 comparison (Bessel order 0 vs. order 1) $\to$ Legendre and Hermite (polynomial termination at integer parameters).